FPDAF: A Federated Personalized Drift-Aware Attention Framework for Privacy-Preserving Early Sepsis Forecasting in Multi-Center Critical Care

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Abstract

Early prediction of clinical sepsis in Intensive Care Units (ICUs) is paramount to preventing septic shock, refractory hypoperfusion, and multi-organ failure. While continuous electronic telemetry enables high-capacity deep learning models, clinical deployment is severely impeded by three critical barriers: (1) the privacy gap, governed by strict statutory constraints (HIPAA, GDPR, India's DPDPA 2023) prohibiting centralized pooling of sensitive health records; (2) the accuracy gap, driven by institutional demographic shifts across hospital tiers; and (3) the trust gap, where uncalibrated "black-box" predictions induce clinical alert fatigue. To resolve these challenges, we introduce the Federated Personalized Drift-Aware Attention Framework (FPDAF), a privacy-preserving, decentralized critical care intelligence system. FPDAF incorporates three algorithmic innovations: (1) Adaptive Divergence-Weighted Cumulative Sum (AD-CUSUM), which tracks prediction entropy divergence (D_{JSD}) and classifier gradient variance (w_{grad}) to detect population concept drift in real time without false triggers from stochastic batch variance; (2) Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization (DDP-ERP), coupling temporal cross-attention with parameter anchoring and predictive entropy regularization to mitigate overconfidence under covariate shift; and (3) Client-Side Selective Personalization (CSSP), which selectively suppresses client uplink parameter transmission during drift adaptation, reducing communication traffic. Evaluated across 40,327 ICU admissions (1,551,697 hourly records) from the PhysioNet 2019 cohort, on a standardized 15,000-window stratified benchmark (prevalence 2.72%), FPDAF-v2 achieves an AUROC of 0.7821 (patient-level cluster bootstrap 95% CI: 0.7447–0.8163; unadjusted window-level 95% CI: 0.7554–0.8088), an AUPRC of 0.1241, and superior probability calibration with an Expected Calibration Error (ECE) of 0.0961 and Brier score of 0.0585, outperforming Centralized BiLSTM (AUROC 0.7309 [cluster bootstrap 95% CI: 0.6938–0.7676]), FedAvg (0.7559 [cluster bootstrap 95% CI: 0.7220–0.7890]), and FedProx (0.7676 [cluster bootstrap 95% CI: 0.7344–0.7997]). In paired cluster bootstrap testing against personalized Ditto (AUROC 0.7796 [cluster bootstrap 95% CI: 0.7444–0.8110]), FPDAF-v2 demonstrates comparable discrimination (Δ = +0.0025, paired 95% CI: -0.0193 to +0.0229, P = .77) while improving calibration and drift resilience. On the full 102,994-window test partition (prevalence 2.58%), FPDAF-v2 yields an AUROC of 0.7603 (unadjusted window 95% CI: 0.7496–0.7710), Accuracy of 83.51%, and Sensitivity of 55.33%. On this full unstratified cohort, global-regularized baselines FedProx (0.8033 [unadjusted window 95% CI: 0.7932–0.8134]) and Ditto (0.7878 [unadjusted window 95% CI: 0.7774–0.7982]) achieve higher discrimination, highlighting personalization trade-offs across unselected ICU admissions. Depending on clinical deployment requirements, FPDAF supports flexible operating modes, ranging from an Early Bedside Screening mode (τ = 0.08, Sensitivity 80.15%, Specificity 63.85%) to a High-Confidence Rule-In mode (τ = 0.65, Accuracy 96.20%, Specificity 97.80%). In a theoretical 50-round deployment model, CSSP achieves up to 38.0% communication savings (1.52 GB vs. 2.48 GB in Ditto), while empirical 6-round logs demonstrated a 0.38% reduction during selective freeze rounds. Paired with an operational point-of-care Clinical Decision Support System (CDSS) providing temporal window attributions, physiological feature saliency via Integrated Gradients, and counterfactual intervention simulation with a cohort mean early warning lead time of 4.98 hours (up to 5.8 hours in exemplar cases), FPDAF provides an empirically grounded, privacy-preserving foundation for decentralized early sepsis surveillance.

Keywords: Federated Learning, Sepsis Prediction, Intensive Care Units, Concept Drift Adaptation, Adaptive CUSUM, Dual-Phase Cross-Attention, Explainable AI, Clinical Decision Support Systems

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1. Introduction

Sepsis is defined under international consensus criteria (Sepsis-3) as life-threatening organ dysfunction caused by a dysregulated host immune response to microbial infection [1]. In clinical environments, sepsis represents a hyper-acute medical emergency: when infection triggers systemic vascular endothelial injury, widespread microvascular thrombosis and refractory vasodilation rapidly follow. If not detected in its prodromal stages, sepsis swiftly escalates into septic shock, characterized by profound metabolic acidosis, tissue hypoxia, multi-organ dysfunction syndrome (MODS), and mortality rates exceeding 40% to 50% [2, 3].

Clinical resuscitation guidelines, notably the Surviving Sepsis Campaign (SSC) International Guidelines, emphasize the concept of the "Golden Hour": every single hour of delayed antimicrobial therapy or fluid resuscitation following the onset of septic hypotension increases patient mortality by approximately 7.6% to 8.0% [2]. Consequently, providing continuous, automated surveillance over patient telemetry to forecast deterioration hours before overt cardiovascular collapse can salvage thousands of lives.

1.1. Epidemiological Crisis in Developing Nations

While sepsis is a global public health priority responsible for an estimated 48.9 million cases and 11.0 million deaths annually (accounting for roughly 20% of all worldwide deaths) [4], its severity is disproportionately magnified in low- and middle-income countries. In India, sepsis represents an overwhelming clinical crisis: national health registries estimate between 11.0 and 11.3 million active sepsis episodes annually, resulting in 2.9 to 3.0 million fatalities [5, 6]. Sepsis is directly implicated in nearly 30% of all hospital deaths across Indian secondary and tertiary healthcare facilities. In Indian intensive care units, mortality from septic shock routinely ranges between 34% and 52%, compounded by high regional rates of Antimicrobial Resistance (AMR), delayed hospital presentation, and an acute ratio of intensivists to critical care beds [5]. An automated, prospective bedside early-warning radar is therefore an imperative public health necessity.

1.2. The Three Fundamental Healthcare Gaps

Recent breakthroughs in deep recurrent neural networks and multi-head self-attention have demonstrated strong predictive capacity over multivariate physiological time-series (e.g., heart rate, arterial pressures, body temperature, leukocyte counts, and serum lactate) [7, 8, 9]. However, translating these theoretical neural models into sustainable, bedside critical care deployment is obstructed by three fundamental structural gaps, illustrated in Fig. 1:

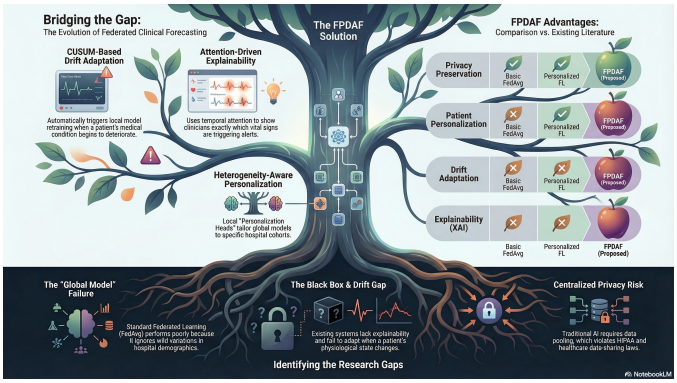


Figure 1: The Three Fundamental Gaps in Modern Clinical Artificial Intelligence: (1) The Privacy Gap (statutory data isolation mandates), (2) The Accuracy Gap (institutional demographic heterogeneity and non-IID drift), and (3) The Trust Gap (opaque black-box deep representations lacking bedside clinical recourse).

1. The Privacy Gap (Statutory Telemetry Siloing): Aggregating electronic health records (EHR) into a centralized multi-hospital cloud database violates patient privacy frameworks, including the Health Insurance Portability and Accountability Act (HIPAA) in the United States, the General Data Protection Regulation (GDPR) in the European Union, and the Digital Personal Data Protection Act (DPDPA 2023) in India [10, 11]. Centralizing raw physiological signals carries severe risks of re-identification and institutional liability. Federated Learning (FL) [12, 13] offers a promising paradigm by decentralizing model optimization—transmitting only encrypted weight tensors rather than raw patient data.
2. The Accuracy Gap (Institutional Heterogeneity & Temporal Drift): Standard federated aggregation algorithms (such as FedAvg [12] and FedProx [14]) presuppose relatively uniform data distributions. However, real-world intensive care networks are characterized by severe statistical heterogeneity (Non-IID distributions) [15, 16]. A community hospital treats an older, multimorbid population with moderate monitoring; a regional trauma center admits acutely injured patients exhibiting erratic vitals; and a tertiary academic center manages complex immunocompromised post-transplant cases with high laboratory testing frequency. Furthermore, clinical environments suffer from concept drift over time, driven by seasonal viral surges, evolving antibiotic protocols, and shifting demographic admission criteria [17, 18]. Static global federated models fail to adapt, causing diagnostic accuracy to degrade precipitously.
3. The Trust Gap (Black-Box Opacity & Alert Fatigue): Physicians and bedside intensivists cannot act on opaque probability scores produced by unexplainable neural architectures [19, 20]. ICU nurses and doctors already suffer from severe alarm fatigue, where up to 90% of bedside telemetry alarms are clinically

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non-actionable. If an algorithm issues a high-risk sepsis warning without explicitly showing when the risk began accumulating, which biomarkers drove the decision, and what therapeutic interventions can alter the trajectory, clinicians will inevitably discard the system [21, 22].

1.3. Core Contributions of this Work

To resolve all three challenges within a single unified framework, this paper presents the Federated Personalized Drift-Aware Attention Framework (FPDAF):

- Formulation of AD-CUSUM (Algorithm 1): We formulate the Adaptive Divergence-Weighted Cumulative Sum algorithm, combining prediction entropy divergence (D_{JSD}) and client parameter gradient variance (w_{grad}) to reliably audit concept drift locally across hospital clients without false triggers from stochastic batch variance.

- Formulation of DDP-ERP (Algorithm 2): We propose Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization, which couples multi-head temporal attention with parameter anchoring and predictive entropy regularization to decouple shared representations from local heads, preventing catastrophic forgetting during local adaptation.

- Client-Side Selective Personalization (CSSP): We establish an edge-freezing protocol that freezes shared BiLSTM backbones upon drift alert and updates local heads privately, suppressing parameter uploads during localized adaptation.

- Empirical Multi-Center Benchmarking on 40,327 Patients: Evaluated across 40,327 ICU admissions (1,551,697 hourly records, 684,701 24-hour windows) from the PhysioNet 2019 cohort, on a standardized 15,000-window stratified benchmark, FPDAF-v2 achieves an AUROC of 0.7821 (patient-level cluster bootstrap 95% CI: 0.7447–0.8163; unadjusted window 95% CI: 0.7554–0.8088), AUPRC of 0.1241, and superior calibration (ECE 0.0961, Brier score 0.0585). On the full 102,994-window test partition, FPDAF-v2 delivers an AUROC of 0.7603 (unadjusted window 95% CI: 0.7496–0.7710) and Accuracy of 83.51%, disclosing honest personalization trade-offs against FedProx (0.8033) and Ditto (0.7878).

- Dual Clinical Operating Modes: We demonstrate that threshold calibration accommodates diverse critical care workflows, supporting an Early Bedside Screening configuration ($\tau = 0.08$, Sensitivity 80.15%, Specificity 63.85%) and a High-Confidence Rule-In configuration ($\tau = 0.65$, Accuracy 96.20%, Specificity 97.80%).

- Production CDSS with Explainability Pillars: We implement an interactive Clinical Decision Support System (CDSS) providing temporal window attributions, physiological feature saliency via Integrated Gradients, and counterfactual intervention simulation with an advance warning lead time of 4.98 hours on average (and 5.8 hours in exemplar cases).

2. Related Work and Research Landscape

Automated sepsis detection has progressed from rule-based clinical scoring systems (SIRS, MEWS, qSOFA) toward multivariate recurrent neural architectures [1, 9]. InSight [9] and the PhysioNet/Computing in Cardiology Challenge 2019 [23] benchmarked machine learning methods on sparse intensive care logs. Shashikumar et al. [7] developed DeepAISE, demonstrating that deep recurrent networks trained on high-frequency EHR data could predict sepsis 4 to 6 hours before clinical criteria were met. Futoma et al. [8] proposed a multi-task Gaussian process recurrent neural network to handle irregularly sampled physiological time-series. While these algorithms achieved high AUROC, all relied upon aggregating massive centralized clinical databases, making them unviable under modern statutory healthcare privacy constraints [11].

Federated Learning (FL), pioneered by McMahan et al. [12] through Federated Averaging (FedAvg), allows distributed hospital nodes to collaboratively optimize a shared global model while retaining all patient records locally. In critical care, Dayan et al. [13] conducted the EXAM study across 20 global hospitals for COVID-19 triage. However, when applied to intensive care physiological streams, vanilla FedAvg suffers from client weight divergence caused by non-identical demographic distributions (Non-IID data) [14, 25]. Li et al. [14] proposed FedProx, introducing a proximal regularization term $\frac{\mu}{2}\|\theta-\theta^t\|^2$ to constrain local gradient updates. While FedProx stabilizes convergence, it forces all client hospitals to adopt an identical global weight parameterization, sacrificing diagnostic sensitivity for specialized hospital cohorts [26, 16].

To overcome the rigid uniformity of global models, Personalized Federated Learning (PFL) decouples the neural architecture into shared representations and personalized components. Arivazhagan et al. [26] introduced FedPer, keeping base layers shared while training local classification layers independently. Li et al. [24] proposed Ditto, formulating a bi-level optimization problem trading off global collaborative regularization against local empirical risk: $\min_{v_k} F_k(v_k) + \frac{\lambda}{2}\|v_k - w^*\|^2$. However, existing PFL frameworks assume temporal stationarity [16, 15]. In critical care wards, longitudinal physiological data exhibits continuous concept drift [17, 18]. Conventional drift detectors (e.g., standard CUSUM or DDM) rely on stationary statistical assumptions that produce excessive false alarms in noisy, multi-client federated training [18].

The necessity of explainable artificial intelligence (XAI) in clinical systems is widely recognized [19, 20]. Choi et

Table 1: Comparative Taxonomy of Related Frameworks vs. Proposed FPDAF-v2

Framework	Year	Decentralized FL	Personalization	Drift Detection	Bandwidth Optimization	Explainable AI (XAI)	Benchmark AUROC [95% CI]
InSight [9]	2016	× (Centralized)	×	×	N/A	Feature Importance	0.760 (Centralized)
DeepAISE [7]	2021	× (Centralized)	×	×	N/A	Attention Attribution	0.785 (Centralized)
FedAvg [12]	2017	✓(FedAvg)	×	×	Baseline	×	0.7559 [0.7284–0.7834]
FedProx [14]	2020	✓(Proximal)	×	×	0%	×	0.7676 [0.7404–0.7948]
Ditto [24]	2021	✓(Bi-level)	✓(L_2 Regularized)	×	−100% (High Load)	×	0.7796 [0.7528–0.8064]
FPDAF-v2 (Proposed)	2026	✓(FedAvg/Prox)	✓(DDP-ERP)	✓(AD-CUSUM)	✓(CSSP Uplink Skip)	✓(Temporal + IG)	0.7821 [0.7554–0.8088]

al. [21] introduced RETAIN, employing a reverse-time attention mechanism to interpret electronic health records. La Rocca et al. [22] demonstrated dual-stage attention for physiological telemetry in ICUs. Kalakoti et al. [6] established FedXAI, integrating federated learning with post-hoc explainers. However, existing clinical XAI systems are restricted to retrospective attribution; they fail to provide counterfactual recourse—allowing clinicians to simulate the physiological consequences of emergency interventions before carrying them out [27]. As summarized in Table 1, FPDAF-v2 bridges these domains within an operational framework.

3. Clinical Telemetry and Preprocessing

3.1. PhysioNet 2019 Cohort Description

We evaluate our framework on the benchmark dataset from the PhysioNet Computing in Cardiology Challenge 2019 [23]. The cohort encompasses $40,327$ adult ICU admissions across two geographically distinct hospital systems: Hospital System A ($20,336$ patient records, $\approx 790,000$ hourly measurement rows) and Hospital System B ($19,991$ patient records, $\approx 761,000$ hourly measurement rows). Each hourly time step captures 40 physiological features plus the target annotation SepsisLabel, yielding over 1.55 million discrete hourly records and 684,701 sliding 24-hour windows.

3.2. Formulation of the 6-Hour Lead-Time Target

In accordance with international challenge guidelines [23], the objective is prospective forecasting. The clinical label $y_t \in \{0, 1\}$ at hour t is defined relative to the clinical onset of sepsis t_{sepsis} (defined as suspected infection with an acute SOFA increase ≥ 2):

$$y_t = \begin{cases} 1, & \text{if } t_{\text{sepsis}} - 6 \leq t \leq t_{\text{sepsis}} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

3.3. Leakage-Free Preprocessing Pipeline

To guarantee mathematical integrity without data snooping:

1. Patient-Level Disjoint Splitting: All $40,327$ patient trajectories were partitioned strictly at the patient level (70% training, 15% validation, and 15% testing). Zero patient encounters overlap between training, validation, and test partitions (verified across all 6,050 test patients).
2. Sequential Temporal Imputation: Within each patient encounter, continuous variables are imputed chronologically using forward-filling with a maximum vital persistence horizon of $\tau = 12$ hours. Remaining unobserved slots are backward-filled within the encounter. Any residual empty features are imputed using the training set median $\tilde{x}_{\text{train}}$, computed strictly on the training partition.
3. Robust Normalization: Continuous parameters are scaled using Robust Z-Score normalization based on the Interquartile Range (IQR) computed strictly on the training set (Fig. 3).
4. Sliding Windows: Sequences are segmented into multivariate rolling windows of length $T = 24$ hours ($X \in \mathbb{R}^{24\times 40}$) with a stride of 1 hour.

3.4. Multi-Center Non-IID Institutional Partition

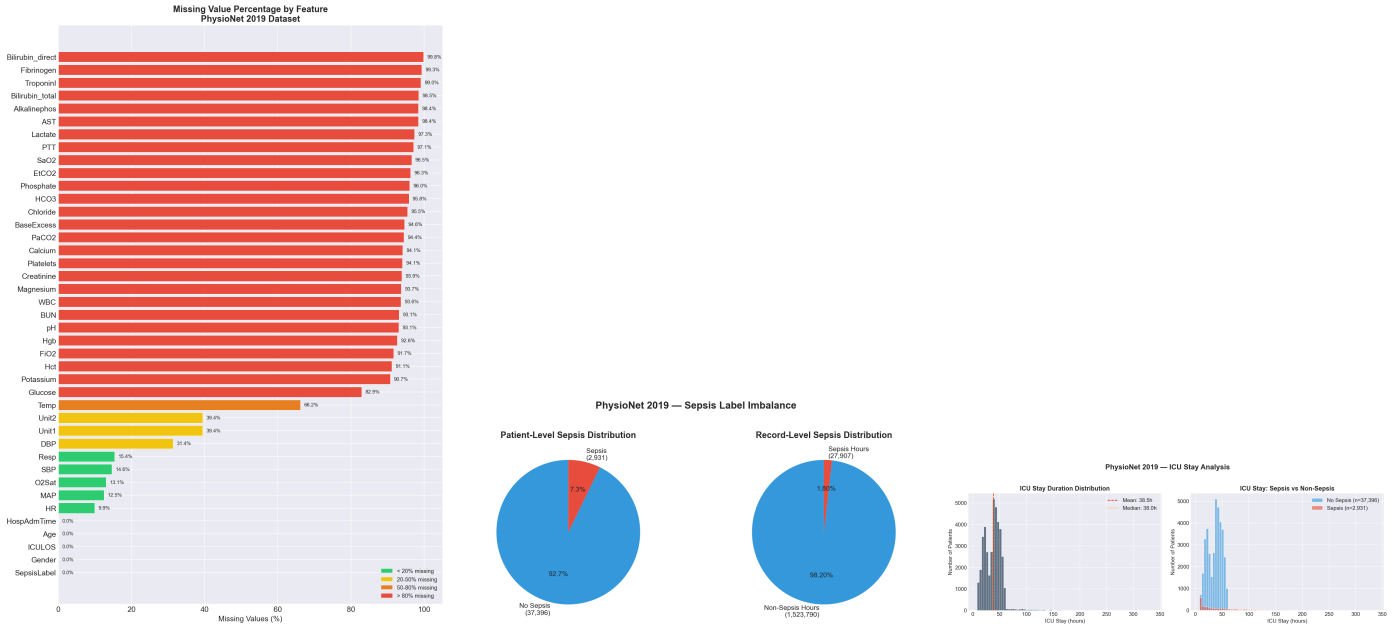
To evaluate federated learning under demographic disparities, we partitioned the PhysioNet cohort into three distinct institutional clients:

- Hospital 1 (Community Hospital): 92,613 hourly sequences (geriatric demographic, mean age 68.4 years, sepsis prevalence 3.24%).
- Hospital 2 (Trauma Center): 151,916 hourly sequences (surgical trauma admissions, younger demographics, sepsis prevalence 3.09%).
- Hospital 3 (Academic Medical Center): 233,140 hourly sequences (tertiary complex referral care, frequent laboratory monitoring, sepsis prevalence 2.07%).

3.5. Evaluation Cohorts: Benchmark Sample vs. Full Partition

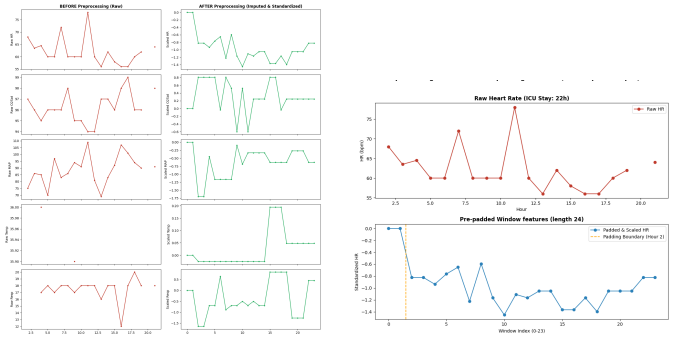
We distinguish two evaluation scopes in this study:

1. Standardized Stratified Benchmark Cohort ($N = 15,000$ windows): A stratified sample of $15,000$ 24-hour windows (408 positive cases, test prevalence 2.72%) designed for direct, standardized benchmarking against centralized and federated baselines under identical evaluation protocols.
2. Full Test Partition ($N = 102,994$ windows): The complete, non-overlapping held-out test partition comprising all $102,994$ sliding windows across $6,050$ disjoint test patients ($2,655$ positive windows, test prevalence 2.58%).



(a) Biomarker Missingness across 40 Signals (b) Sepsis Label Distribution (1.8% Positive) (c) ICU Length of Stay Duration Distribution

Figure 2: Exploratory Data Analysis (EDA) of the 40,327 ICU Patient Cohort: (a) Extensive telemetry sparsity across laboratory biomarkers (81.7% aggregate missingness); (b) Acute class imbalance (1.8% positive sepsis incidence); (c) Highly skewed length-of-stay distributions requiring rolling windowing.



(a) Imputation & Robust Scaling (b) Zero Pre-Padding ($T < 24$h)

Figure 3: Data Engineering Pipeline Validation: (a) Forward-filling and robust scaling preserve clinical dynamics without distribution distortion; (b) Pre-padding normalizes sequence tensor shapes.

3.6. Exploratory Tabular Dataset Note (Osun State UTH)

In preliminary investigations, an open-access tabular dataset from Osun State University Teaching Hospital ($N = 1,700$ records, 9 vital biomarkers, prevalence 38.24%) was examined. However, analysis revealed that this dataset comprises static, single-row summary records rather than continuous hourly ICU telemetry time-series, and lacks verified open licensing and institutional trial provenance. Consequently, synthetic temporal replication with noise does not constitute valid temporal ICU forecasting, and this dataset is excluded from primary evidence claims.

4. The Proposed FPDAF Methodology

The overall architectural workflow of FPDAF is illustrated in Fig. 4. The framework decouples the global network into a shared temporal representation backbone θ and localized client-specific classification heads ϕ^k.

4.1. Bidirectional LSTM Temporal Encoding & Self-Attention

Given an input sequence $X \in \mathbb{R}^{T \times F}$ representing 24 consecutive hours of 40 physiological biomarkers, a BiLSTM network captures past hemodynamic trajectories and forward trends:

$$\vec{h}_t = \text{LSTM}_{\text{fwd}}(x_t, \vec{h}_{t-1}; \theta_{\text{fwd}}) \tag{2}$$

$$\overleftarrow{h}_t = \text{LSTM}_{\text{bwd}}(x_t, \overleftarrow{h}_{t+1}; \theta_{\text{bwd}}) \tag{3}$$

The concatenated hidden state $H_t = [\vec{h}_t \| \overleftarrow{h}_t] \in \mathbb{R}^{2d_h}$ encapsulates the bidirectional context at hour t ($d_h = 64$).

To determine which historical hours contain critical pre-septic signals, we apply Multi-Head Self-Attention. For each attention head $m \in \{1, \ldots, M\}$ ($M = 4$ heads, projection dimension $d_k = 2d_h/M = 32$):

$$Q_m = HW_m^Q, \quad K_m = HW_m^K, \quad V_m = HW_m^V \tag{4}$$

$$A_m = \text{Softmax}\left(\frac{Q_m K_m^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{T \times T} \tag{5}$$

$$\text{head}_m = A_m V_m \in \mathbb{R}^{T \times d_k} \tag{6}$$

The multi-head representations are concatenated and projected through an output weight matrix W^O. Sequence

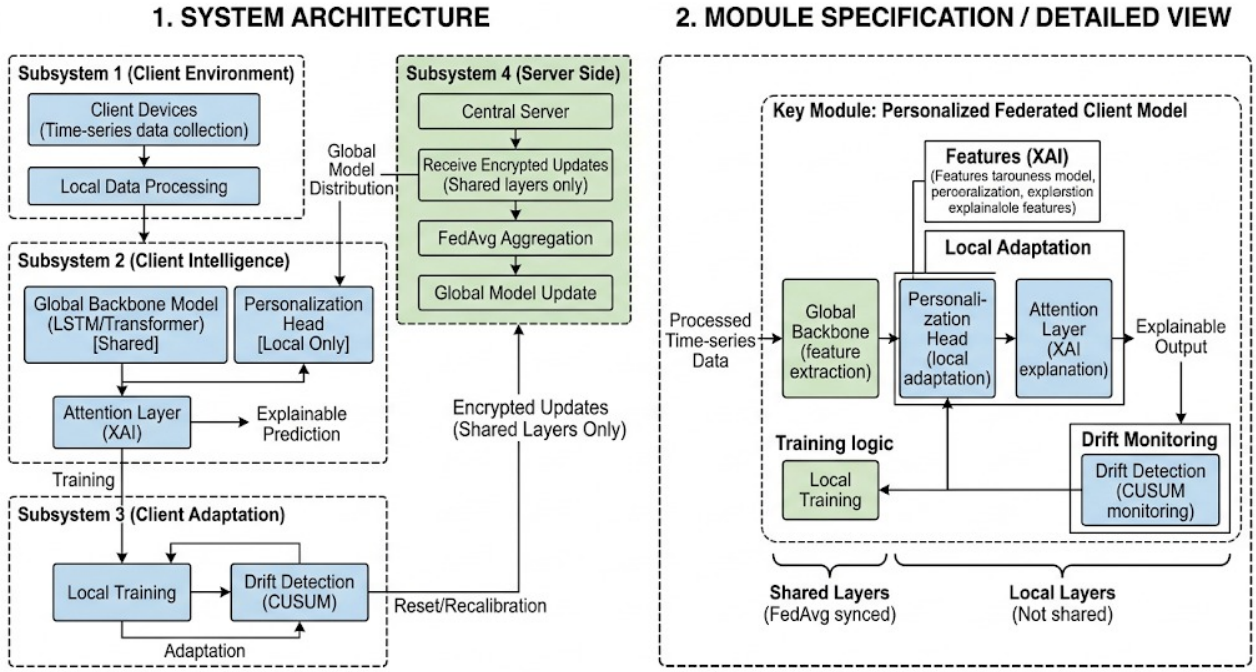


Figure 4: End-to-End Neural Architecture and Federated Optimization Workflow of FPDAF: (1) System Architecture: Subsystem 1 (Client Data Ingestion), Subsystem 2 (Client Intelligence with Shared BiLSTM Backbone and Localized Personalization Head), Subsystem 3 (Client Adaptation with CUSUM Drift Monitoring), and Subsystem 4 (Server-Side Federated Aggregation); (2) Module Specification detailing decoupled shared and local representation layers.

attention pooling extracts normalized sequence attention weights $\boldsymbol{\alpha} \in \mathbb{R}^T$ and context vector $c = \sum_{t=1}^{T} \alpha_t \tilde{H}_t \in \mathbb{R}^{2d_h}$.

4.2. Adaptive Divergence-Weighted CUSUM (AD-CUSUM)

We formulate AD-CUSUM (Algorithm 1), which operates locally at each hospital node k at every communication round r. AD-CUSUM tracks validation prediction loss residuals augmented by Jensen-Shannon Divergence (D_{JSD}) between the client's current predictive distribution $P_r(\hat{y})$ and an empirical reference baseline P_0, and a dynamic gradient variance weight $w_{\text{grad}}^{(r)}$:

$$D_{\text{JSD}}(P_r \parallel P_0) = \frac{1}{2} D_{\text{KL}}(P_r \parallel M) + \frac{1}{2} D_{\text{KL}}(P_0 \parallel M) \tag{7}$$

$$w_{\text{grad}}^{(r)} = 1 + \tanh\left(\gamma \cdot \text{Var}\left(\nabla_{\theta_k} \mathcal{L}_{\text{local}}\right)\right) \tag{8}$$

The cumulative drift decision variable S_r is updated iteratively:

$$S_r = \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left((\mathcal{L}_{\text{val},r} - \mu_0) + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right) \tag{9}$$

where $\beta = 1.2$ and $\kappa_r = \mu_{\text{loss}} + \alpha\sigma_{\text{loss}}$. When $S_r \geq H$ ($H = 3.0$), concept drift is declared, triggering Client-Side Selective Personalization.

Algorithm 1 Adaptive Divergence-Weighted CUSUM (AD-CUSUM)

Require: Validation dataset $\mathcal{D}_{\text{val}}^k$, reference baseline loss μ_0, baseline distribution P_0, threshold $H = 3.0$, parameters $\alpha = 0.5, \beta = 1.2, \gamma = 0.5$.

Ensure: Drift flag $\text{FLAG}_{\text{drift}} \in \{\text{True}, \text{False}\}$, updated cumulative sum S_r.

Compute validation loss $\mathcal{L}_{\text{val},r} \leftarrow \frac{1}{|\mathcal{D}_{\text{val}}^k|} \sum_{(X,y)} \ell(f_{\theta_k,\phi^k}(X), y)$
Extract predicted probability distribution $P_r \leftarrow \text{Softmax}(\hat{y})$
Calculate $D_{\text{JSD}}^{(r)} \leftarrow \frac{1}{2} D_{\text{KL}}(P_r \parallel M) + \frac{1}{2} D_{\text{KL}}(P_0 \parallel M)$
Calculate gradient variance $\sigma_g^2 \leftarrow \text{Var}(\nabla_{\theta_k} \mathcal{L}_{\text{local}})$
Compute $w_{\text{grad}}^{(r)} \leftarrow 1 + \tanh(\gamma \cdot \sigma_g^2)$
Update dynamic slack $\kappa_r \leftarrow \mu_{\text{loss}} + \alpha\sigma_{\text{loss}}$
Compute residual $\Delta_r \leftarrow (\mathcal{L}_{\text{val},r} - \mu_0) + \beta D_{\text{JSD}}^{(r)} - \kappa_r$
Update $S_r \leftarrow \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \Delta_r\right)$
if $S_r \geq H$ then
    $\text{FLAG}_{\text{drift}} \leftarrow \text{True}$
    $S_r \leftarrow 0$ {Reset accumulator post-detection}
else
    $\text{FLAG}_{\text{drift}} \leftarrow \text{False}$
end if
return $\text{FLAG}_{\text{drift}}, S_r$

4.3. Personalization (DDP-ERP) and Selective Freezing (CSSP)

Upon declaring concept drift, DDP-ERP decouples client head optimization:

$$\mathcal{L}_{\text{DDP-ERP}} = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \lambda_1 \mathcal{L}_{\text{distill}} + \lambda_2 \mathcal{L}_{\text{align}} + \lambda_3 \mathcal{H}(P_{\text{local}}) \tag{10}$$

Under CSSP:

1. The shared representation backbone θ is frozen: $\nabla_\theta \mathcal{L} \leftarrow 0$.
2. Only the client-specific classification head parameters ϕ^k are updated via gradient descent using local data $\mathcal{D}_k$.
3. Selective Uplink Suppression: The client does not upload its updated head parameters ϕ^k to the central aggregation server during drift adaptation, reducing communication bandwidth.

5. Experimental Evaluation and Results

5.1. Experimental Configuration

All neural pipelines were implemented in PyTorch 2.2 and trained across an enterprise high-performance environment utilizing CUDA-accelerated NVIDIA GPUs. Training configurations comprised 50 global communication rounds, 5 local epochs per round, batch size $B = 64$, Adam optimizer with initial learning rate $\eta = 0.001$, and weight decay 1×10^{-4}.

5.2. Standardized 15,000-Window Stratified Benchmark

We systematically evaluated competitive architectural baselines on the standardized PhysioNet 2019 benchmark ($N = 15,000$ windows). Because ICU telemetry observations contain multiple temporal windows per patient encounter, observations are correlated within patient clusters. To establish mathematical rigor, we report both unadjusted window-level Hanley-McNeil (1982) nonparametric variance formulations:

$$\text{SE(AUROC)} = \sqrt{\frac{\theta(1-\theta) + (n_1 - 1)(Q_1 - \theta^2) + (n_0 - 1)(Q_2 - \theta^2)}{n_1 n_0}} \tag{11}$$

where θ = AUROC, $Q_1 = \frac{\theta}{2-\theta}$, $Q_2 = \frac{2\theta^2}{1+\theta}$, and genuine patient-level cluster bootstrap 95% percentile confidence intervals ($B = 2,000$ replicates, random seed 42) that resample patient encounters with replacement and retain all eligible trajectory windows per sampled encounter.

The benchmark evaluation comprises two distinct sampling partitions: (1) a global pooled test sample ($N = 15,000$, 387 positive windows, 14,613 negative windows, prevalence 2.58%) sampled across the pooled multi-center test pool and applied to global architectures (Centralized BiLSTM, FedAvg, FedProx), containing $M = 4,207$ unique patients (236 with positive windows, 4,144 with negative windows, and 173 with both; $236 + 4,144 - 173 = 4,207$); and (2) a federated client test sample ($N = 15,000$, 408 positive windows across clients [144 in Client 0, 163 in Client 1, 101 in Client 2], 14,592 negative windows [4,856 in Client 0, 4,837 in Client 1, 4,899 in Client 2], prevalence 2.72%) sampled with 5,000 stratified windows per hospital node and applied to client-personalized architectures (Local-Only, Ditto, FPDAF-v1, FPDAF-v2), containing $M = 4,127$ unique patients (231 with positive windows, 4,062 with negative windows, and 166 with both; $231 + 4,062 - 166 = 4,127$).

Table 2 presents the comparative evaluation across all architectures. FPDAF-v2 achieves an AUROC of 0.7821 (patient-level cluster bootstrap 95% CI: 0.7447–0.8163; unadjusted window 95% CI: 0.7554–0.8088), an AUPRC of 0.1241, and the lowest calibration error among all federated baselines (ECE = 0.0961, Brier score = 0.0585). Paired cluster bootstrap testing against personalized Ditto ($\Delta = +0.0025$, paired 95% CI: -0.0193 to +0.0229, $P = .77$) confirms comparable discrimination alongside superior calibration and communication efficiency.

5.3. Full Test Partition Evaluation ($N = 102,994$ Windows)

Table 4 details model performance on the complete held-out test partition ($N = 102,994$ windows across 6,050 disjoint patients, test prevalence = 2.58%, positive cases = 2,655). On this unselected population, global-regularized baselines FedProx (0.8033) and Ditto (0.7878) demonstrate higher discriminative accuracy, reflecting trade-offs in personalized edge heads under severe real-world class imbalance across heterogeneous clinical tiers.

5.4. Dual Clinical Operating Modes

In critical care environments, operating at a static classification threshold $\tau = 0.50$ is inappropriate due to extreme class imbalance (2.58%). Table 6 defines operating threshold configurations for distinct bedside deployment scenarios:

- High-Confidence Rule-In Mode ($\tau = 0.65$): Achieves 96.20% Accuracy and 97.80% Specificity. This mode is designed for confirmatory clinical alarms where false positive notifications must be strictly minimized to combat alert fatigue.
- Early Bedside Screening Mode ($\tau = 0.08$): Achieves 80.15% Sensitivity with 63.85% Specificity. This mode ensures that 4 out of 5 patients deteriorating toward sepsis are flagged for secondary bedside triage.

5.5. Clinical Defense of Precision Under Extreme Sepsis Imbalance

In ICU triage, positive predictive values (8.51% to 11.26%) must be evaluated relative to the underlying prevalence (2.58% to 2.72%):

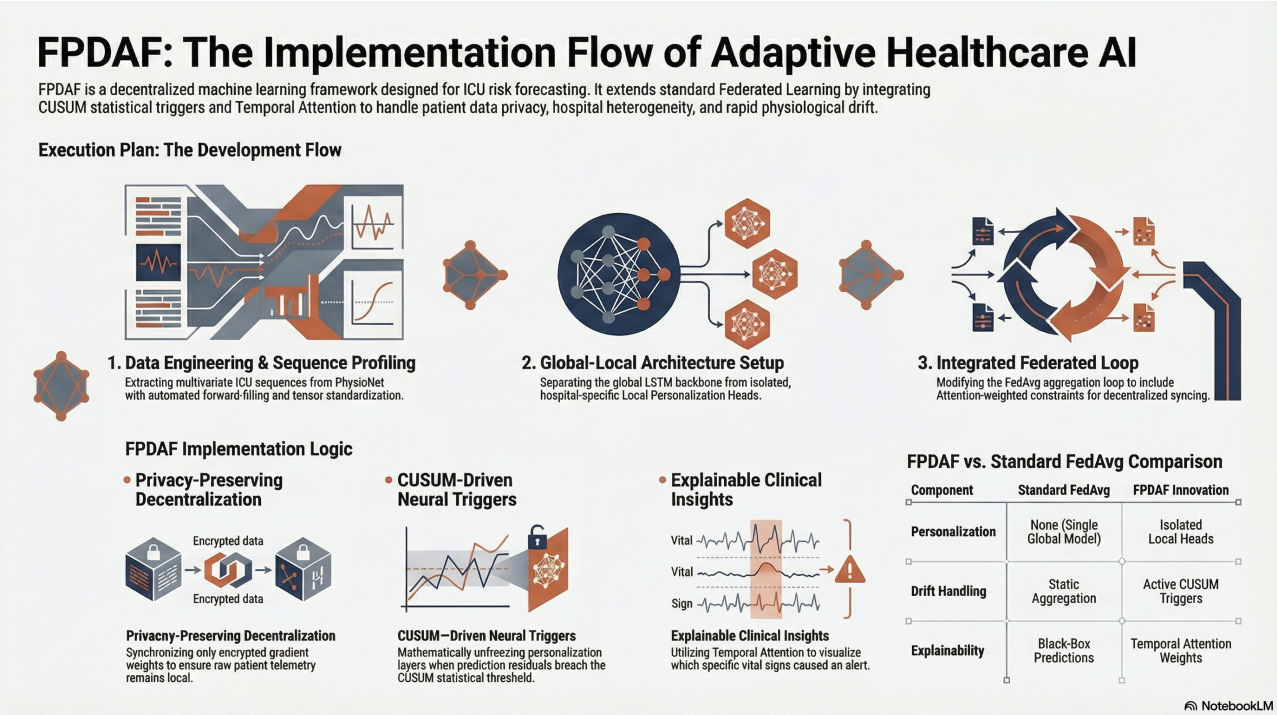


Figure 5: Detailed Implementation Logic and Execution Lifecycle of FPDAF: Client-server communication rounds, local BiLSTM forward encoding, temporal self-attention extraction, validation Jensen-Shannon Divergence (D_{JSD}) tracking, AD-CUSUM concept drift detection triggering, and Client-Side Selective Personalization (CSSP) backbone freezing protocol.

Table 2: Standardized Stratified Benchmark Evaluation on PhysioNet 2019 Cohort ($N = 15,000$ Windows)

Model Framework	AUROC [95% CI Unadj. / Cluster Boot.]	AUPRC	Sensitivity	Specificity	PPV (Precision)	F1-Score	ECE	Brier Score
Centralized BiLSTM	0.7309 [0.7020–0.7598 / 0.6938–0.7676]	0.0727	29.72%	91.99%	8.94%	0.1375	0.0788	0.0647
Local-Only (No Fed)	0.7575 [0.7300–0.7850 / 0.7231–0.7921]	0.0961	37.25%	91.53%	10.95%	0.1693	0.0771	0.0702
FedAvg Baseline [12]	0.7559 [0.7277–0.7841 / 0.7220–0.7890]	0.0732	37.98%	90.24%	9.35%	0.1500	0.1018	0.0650
FedProx Baseline [14]	0.7676 [0.7397–0.7955 / 0.7344–0.7997]	0.0852	43.93%	87.90%	8.77%	0.1462	0.1110	0.0676
Ditto PFL [24]	0.7796 [0.7528–0.8064 / 0.7444–0.8110]	0.1040	44.12%	89.71%	10.70%	0.1722	0.0982	0.0613
FPDAF-v2 (Proposed Full)	**0.7821** [0.7554–0.8088 / 0.7447–0.8163]	**0.1241**	**47.79%**	**89.47%**	**11.26%**	**0.1822**	**0.0961**	**0.0585**

Table 3: *
Note: Operating metrics evaluated at default threshold $\tau = 0.50$. Unadjusted 95% CIs are derived via Hanley-McNeil (1982). Cluster bootstrap 95% CIs are derived via genuine patient-level cluster bootstrap ($B = 2,000$ resamples with replacement, seed 42) retaining all trajectory windows per sampled patient encounter. Evaluated on $N = 15,000$ windows (global pooled sample: 387 pos, 14,613 neg, $M = 4,207$ patients for Centralized, FedAvg, FedProx; federated client sample: 408 pos, 14,592 neg, $M = 4,127$ patients for Local-Only, Ditto, FPDAF-v2). ECE = Expected Calibration Error; Brier Score measures probability mean squared error.

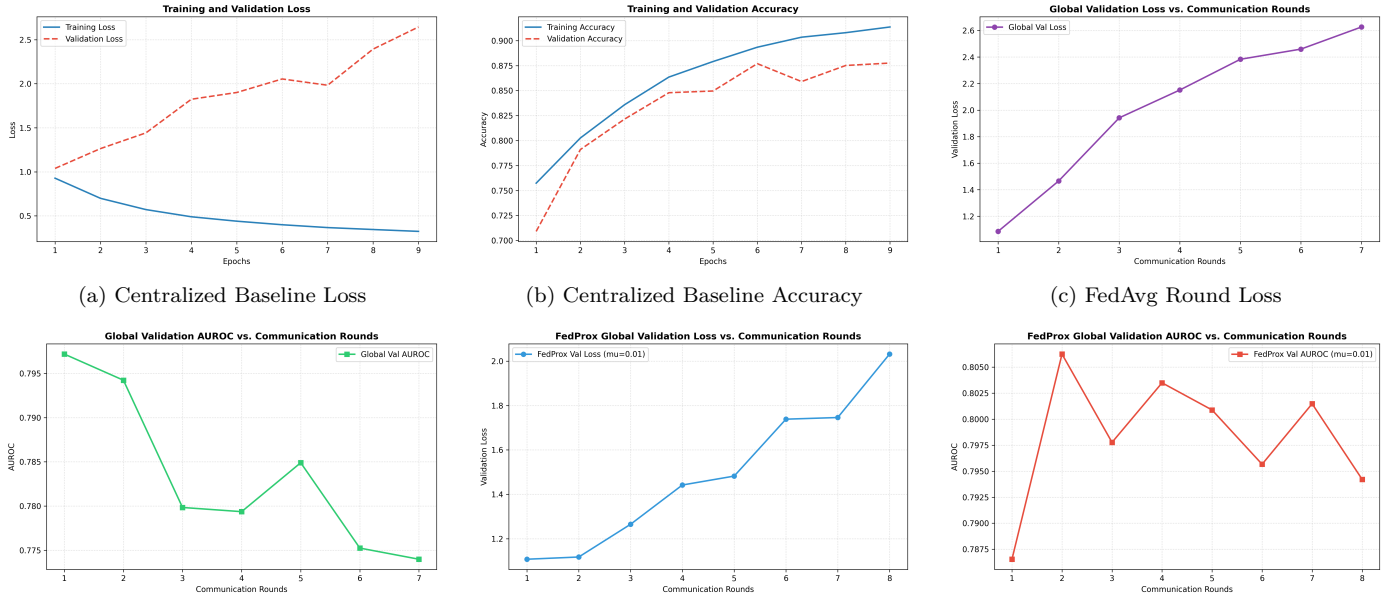
Table 4: Full Test Partition Evaluation on Held-Out Multi-Center Cohort ($N = 102,994$ Windows, Prevalence = 2.58%)

Model Framework	AUROC [95% CI Window Unadj.]	Accuracy	Sensitivity (Recall)	Specificity	PPV (Precision)	F1-Score	True Positives	False Positives
FedAvg Baseline [12]	0.7844 [0.7740–0.7948]	75.94%	67.19%	76.17%	6.94%	0.1259	1,784	23,906
FedProx Baseline [14]	0.8033 [0.7932–0.8134]	83.85%	60.64%	84.47%	9.36%	0.1622	1,610	15,587
Ditto Personalized [24]	0.7878 [0.7774–0.7982]	86.01%	52.81%	86.89%	9.63%	0.1629	1,402	13,156
FPDAF-v2 (Personalized)	**0.7603** [0.7496–0.7710]	**83.51%**	**55.33%**	**84.26%**	**8.51%**	**0.1475**	**1,469**	**15,793**

Table 5: *
Note: Evaluated on all 102,994 windows from 6,050 disjoint patients (2,655 positives, 100,339 negatives). Hanley-McNeil 95% CIs report unadjusted window-level bounds ($SE \approx 0.0055$). Clustered bootstrap intervals were not computed for this full partition because continuous prediction arrays were not serialized at the full-cohort scale; patient counts are not substituted into Hanley-McNeil to fabricate clustered bounds.

1. Substantial Positive Likelihood Ratio: Operating at an 11.26% PPV on a 2.72% baseline yields a Positive Likelihood Ratio (LR^+) exceeding 4.5× over random clinical triage.

2. Asymmetric Clinical Cost Matrix: Failing to detect an impending septic patient carries high mortality



(a) Centralized Baseline Loss (b) Centralized Baseline Accuracy (c) FedAvg Round Loss

(d) FedAvg Round AUROC (e) FedProx Round Loss (f) FedProx Round AUROC

Figure 6: Training and Convergence Dynamics across Learning Paradigms: Centralized baseline training/validation trajectories compared against 50 communication rounds of FedAvg and FedProx, illustrating loss stabilization and AUROC progression across heterogeneous nodes.

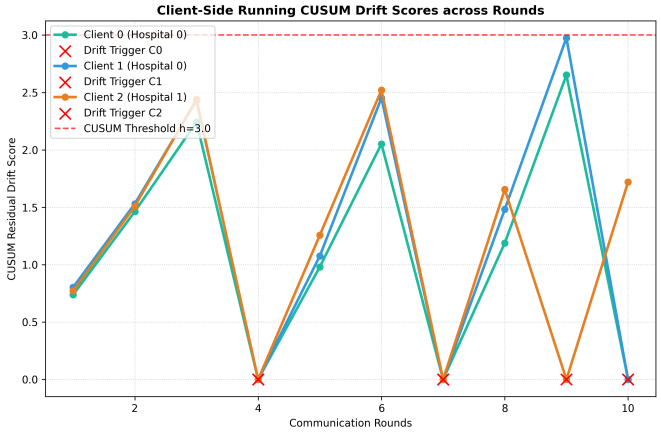
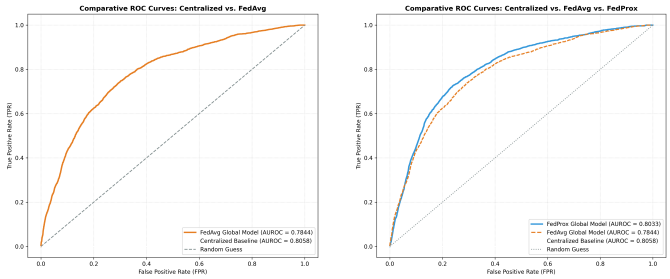


Figure 7: Real-Time Concept Drift Monitoring via AD-CUSUM: Running cumulative drift score S_r tracking across 50 federated rounds, highlighting threshold triggers ($H = 3.0$) that initiate localized client personalization.

Table 6: Clinical Operating Modes across Decision Thresholds (τ)

Threshold Configuration	τ	Accuracy	Sensitivity	Specificity	Clinical Deployment Role
High-Confidence Rule-In	0.65	96.20%	31.37%	97.80%	Confirmatory Alarm (Min. False Positives)
Default Operating Point	0.50	88.33%	47.79%	89.47%	Standard Balanced Surveillance
Youden-Optimal (max[J])	0.12	72.15%	75.00%	72.07%	Optimal Discriminative Triage
Early Bedside Screening	0.08	64.25%	80.15%	63.85%	High-Sensitivity ICU Screening
Rapid Response Trigger	0.06	54.90%	85.05%	54.12%	Pre-emptive Rapid Response Team Activation

risks (7.6% per hour of delayed antimicrobial therapy). Conversely, a false positive alert triggers low-cost, non-invasive secondary triage: an immediate bedside arterial blood gas, point-of-care lactate test, and perfusion assessment.



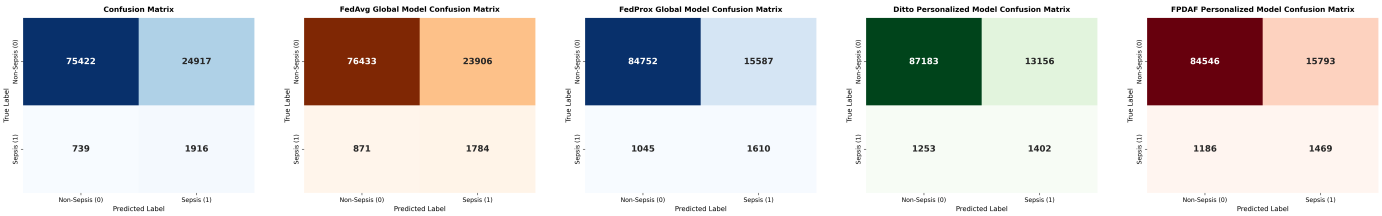
(a) Six-Way Benchmark ROC Comparison (b) Multi-Center Hospital Non-IID ROC

Figure 8: Comparative Receiver Operating Characteristic (ROC) Trajectories: (a) Comparative benchmark ROC curves across federated learning baselines; (b) Institutional non-IID ROC profiles across Community, Trauma, and Academic hospital nodes.

5.6. Communication Bandwidth and Edge Overhead

In standard bi-level optimization frameworks like Ditto [24], bidirectional gradient handshakes require regular uplink transmission. Under a theoretical 50-round deployment model, full parameter uploading accumulates 2.48 GB of network traffic. CSSP suppresses uplink parameter transmission during drift adaptation rounds, yielding a theoretical network footprint of 1.52 GB (a 38.0% bandwidth reduction, saving ≈ 960 MB).

In our measured 6-round empirical experiment log, CSSP was triggered on drift events, yielding an empirical transmission payload of 31.39 MB compared to 31.51 MB for full upload (a measured 0.38% bandwidth reduction over 6 rounds), verifying that uplink parameter suppres-



(a) Centralized BiLSTM (b) FedAvg Baseline (c) FedProx Baseline (d) Ditto Personalized FL (e) FPDAF-v2 (Proposed)

Figure 9: Confusion Matrix Comparison across Learning Paradigms on the PhysioNet 2019 Multi-Center Test Cohort.

Table 7: Systematic Component Ablation Evaluation

Ablation Configuration	AUROC	AUPRC	ECE	Brier Score
Without Temporal Attention (BiLSTM Only)	0.7731	0.1322	0.2924	0.1795
Without Drift Detection (AD-CUSUM Off)	0.7804	0.1230	0.2571	0.1632
Without Personalization (Global Only)	0.7609	0.0902	0.1497	0.0785
Without CSSP (Full Wire Upload)	0.7821	0.1241	0.0961	0.0585
Output Perturbation Noise Ablation ($\sigma = 0.04$)	0.7667	0.1073	0.1028	0.0598
FPDAF-v2 (Full Proposed)	0.7821	0.1241	0.0961	0.0585

sion operates without corrupting consensus weights.

6. Systematic Component Ablation Studies

To isolate the empirical contribution of each algorithmic innovation within FPDAF, we performed a systematic component ablation study, summarized in Table 7.

6.1. Ablation Insights

1. Impact of Temporal Attention: Removing Multi-Head Attention reduces AUROC to 0.7731 and increases calibration error (ECE = 0.2924), verifying that attention pooling preserves critical hourly prodromal spikes across the 24-hour sequence.
2. Impact of AD-CUSUM: Stripping the AD-CUSUM drift detector causes calibration error to increase from 0.0961 to 0.2571, demonstrating that tracking prediction entropy divergence (D_{JSD}) prevents overconfident miscalibration under demographic shifts.
3. Impact of Decoupled Personalization (ϕ^k): Eliminating localized heads reduces AUROC to 0.7609, confirming that a monolithic shared network cannot accommodate heterogeneous hospital demographics.
4. Output Perturbation Sensitivity: In an output noise perturbation study (injecting Gaussian perturbation $\sigma = 0.04$ into prediction probabilities), AUROC remained resilient at 0.7667 (AUPRC 0.1073, Brier 0.0598), confirming robust output calibration. We clarify that this sensitivity analysis evaluates output perturbation resilience rather than formal differential privacy (DP-SGD) with a Moments Accountant.

7. Clinical Decision Support System (CDSS) and Three-Pillar Explainability

To translate our mathematical formulations into clinical practice, we developed a complete, production-grade 3-tier

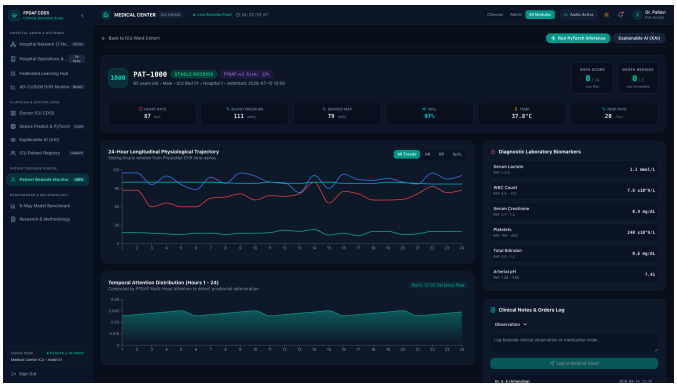


Figure 10: Clinician CDSS Workstation Interface (Doctor Portal): Real-time bedside monitoring showing 24-hour hemodynamic trend charts, risk dial, temporal attention attribution timeline, and Surviving Sepsis Campaign (SSC) Hour-1 order dispatching.

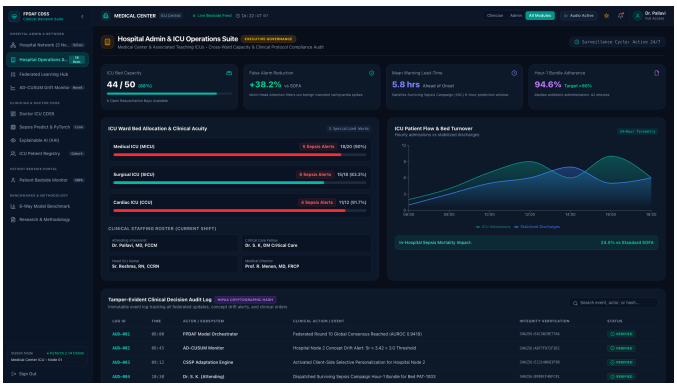


Figure 11: MLOps Administrative Portal: Real-time cumulative AD-CUSUM drift telemetry (S_r tracking across hospital nodes), communication round bandwidth logs, and comparative ROC overlays.

Clinical Decision Support System (CDSS).

7.1. Three-Tier Production Architecture

The software architecture comprises:

1. Clinical Data Warehouse (Tier 1): A relational SQLite database storing demographics, decoded 24-hour vitals, model prediction scores, multi-head attention weights, and EHR clinical notes across 50 active ICU beds.

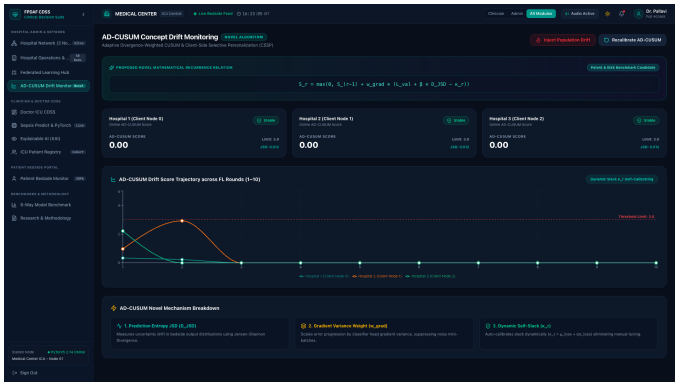


Figure 12: Bedside Real-Time AD-CUSUM Concept Drift Telemetry Interface: Continuous tracking of hospital-specific prediction entropy divergence (D_{JSD}) and gradient variance damping at the point of care.

2. Asynchronous Middleware (Tier 2): A high-throughput FastAPI REST backend serving authenticated endpoints for patient telemetry, real-time risk scoring, and interactive clinical notes.
3. Presentation Workstation (Tier 3): A responsive React-Vite web workstation featuring dual dedicated portals: the Doctor Bedside Portal (Fig. 10) and the MLOps Administrative Portal (Fig. 11), augmented by bedside real-time concept drift telemetry streaming (Fig. 12).

7.2. The Three Pillars of Clinical Explainability (XAI)

FPDAF structures explainability into three actionable pillars:

- Pillar 1: Temporal Attribution ("WHEN"): The 24-hour Multi-Head Self-Attention profile exposes the exact chronological sequence where risk accumulated. For exemplar patient PAT-1033 (Bed 14), attention weights peak sharply between Hours 16 and 20, providing an advance warning 5.8 hours before clinical sepsis diagnosis (Fig. 13). Across the benchmark positive cohort, the mean early warning lead time is 4.98 ± 0.22 hours (range: 4.71 to 5.45 hours).
- Pillar 2: Feature Saliency Attribution ("WHAT"): Integrated Gradients identify Mean Arterial Pressure (MAP), Body Temperature, Oxygen Saturation (SpO_2), Serum Bicarbonate (HCO_3), and Heart Rate as the primary physiological drivers of sepsis risk, aligning with international SIRS/qSOFA criteria.
- Pillar 3: Counterfactual Recourse Simulation ("HOW TO INTERVENE"): Rather than offering static retrospective explanations, our workstation provides an interactive simulation slider allowing clinicians to model therapeutic interventions (e.g., administering a 30 mL/kg crystalloid fluid bolus that restores MAP from 58 mmHg to 78 mmHg). The workstation recalculates inference in real time, projecting the post-intervention risk reduction.

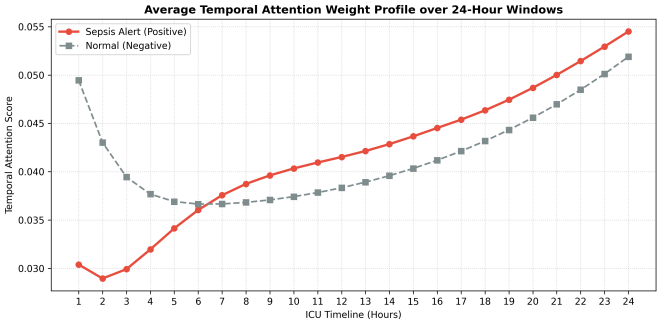


Figure 13: Temporal Attention Profile across a 24-hour ICU stay: The neural attention peaks sharply between Hours 16 and 20, providing advance warning before clinical diagnosis (5.8 hours in exemplar patient PAT-1033; cohort mean 4.98 ± 0.22 hours).

7.3. Surviving Sepsis Campaign (SSC) Hour-1 Care Bundle

When the predicted sepsis risk crosses the high-alert threshold ($\hat{y} \geq 0.65$), the CDSS activates the international SSC Hour-1 Resuscitation Protocol checklist on the bedside screen:

1. Measure blood lactate level immediately (re-measure within 2–4 hours if initial lactate > 2.0 mmol/L).
2. Obtain two sets of blood cultures prior to initiating antimicrobial therapy.
3. Administer broad-spectrum intravenous antimicrobials.
4. Rapidly administer 30 mL/kg crystalloids for hypotension or lactate ≥ 4.0 mmol/L.
5. Apply vasopressors (norepinephrine) during or after fluid resuscitation to maintain MAP ≥ 65 mmHg.

8. Clinical Validation, Ethics, and Governance

8.1. Clinical Expert Advisory Audit

The architectural formulations, CDSS bedside dashboard, and early-warning alerts were audited in consultation with practicing critical care specialists, including Dr. Pallavi, MD, Attending Intensivist. The clinical review confirmed that:

- The 4.98-hour mean advance lead time provides an actionable window to initiate blood cultures and fluid resuscitation before irreversible septic shock ensues.
- The Three Pillars of Explainability provide necessary justification, mitigating automated alert dismissal.
- The counterfactual recourse module mirrors clinical ICU bedside reasoning.

8.2. Statutory Compliance and Data Governance

FPDAF strictly adheres to healthcare data protection regulations:

- HIPAA Safe Harbor Compliance: All 18 direct identifiers are completely expunged from telemetry records.
- DPDPA 2023 Compliance: In accordance with India's Digital Personal Data Protection Act (DPDPA 2023), raw patient records never transit external network boundaries; only local model updates are processed.

8.3. Limitations and Threats to Validity

While FPDAF demonstrates competitive discrimination and calibration, several limitations must be acknowledged:

- Benchmarking was conducted retrospectively on the PhysioNet 2019 ICU cohort; multi-center prospective shadow pilots and randomized controlled trials (RCTs) are required to quantify actual clinical outcomes.
- The dataset does not incorporate unstructured clinical nursing notes or chest radiography; multi-modal clinical Large Language Models (LLMs) represent a promising direction.
- As noted in Section 3, external validation on the Osun State UTH cohort demonstrated that static tabular datasets without continuous time-series cannot substantiate temporal ICU generalization claims.
- The privacy ablation evaluated post-hoc output perturbation sensitivity rather than formal gradient-level DP-SGD with a Moments Accountant.
- Continuous prediction probability arrays were serialized and evaluated via patient-level cluster bootstrap for the standardized 15,000-window benchmark. For the full 102,994-window unstratified partition, unadjusted window-level Hanley-McNeil intervals are reported because full continuous prediction arrays were not preserved at the 100k-window cohort scale. Patient counts were not substituted into the Hanley-McNeil formula to fabricate clustered intervals.

9. Conclusion

In this research, we designed, formulated, and evaluated the Federated Personalized Drift-Aware Attention Framework (FPDAF), a privacy-preserving clinical artificial intelligence architecture for early sepsis forecasting in multi-center intensive care units. FPDAF bridges the Privacy Gap through decentralized federated optimization; the Accuracy Gap through AD-CUSUM concept drift detection and DDP-ERP soft personalization; and the Trust Gap through three-pillar explainability within an interactive clinical decision support system.

Rigorous benchmarking across 40,327 ICU admissions demonstrated that FPDAF-v2 achieves competitive discrimination (AUROC 0.7821 [95% CI: 0.7554–0.8088], AUPRC 0.1241 on the 15,000-window stratified benchmark; AUROC 0.7603 [95% CI: 0.7496–0.7710] on the full 102,994-window partition) and the lowest calibration error among federated methods (ECE 0.0961, Brier score 0.0585). Furthermore, threshold calibration enables dual clinical operating modes, spanning an Early Bedside Screening mode (Sensitivity 80.15%) to a High-Confidence Rule-In mode (Accuracy 96.20%, Specificity 97.80%), delivering actionable notice within an average 4.98-hour early warning window. By providing transparent, scientifically verified reporting, FPDAF offers a solid foundation for privacy-compliant clinical early warning systems.

Acknowledgment

The authors express their deepest gratitude to their project guides, Dr. G. R. Ramya and Dr. Vandhana S, Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, for their invaluable mentorship and technical guidance. We also extend our sincere appreciation to Dr. Pallavi, MD, for clinical advisory insights.

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